

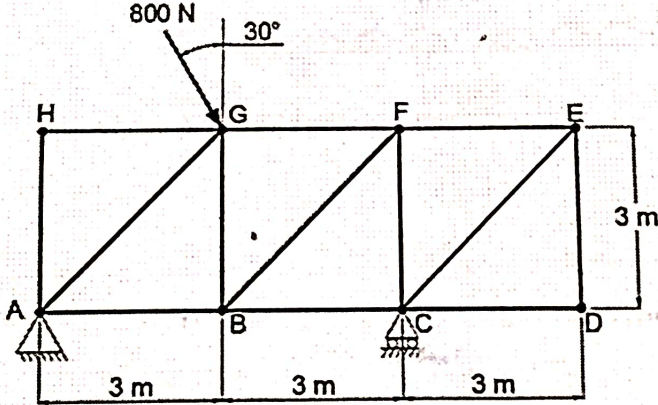
STRUCTURAL ANALYSIS – I
CEC 401

FM: 25

TIME: 90 MINUTES

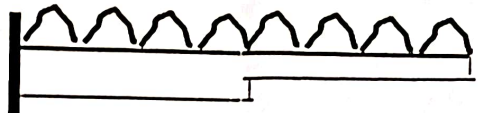
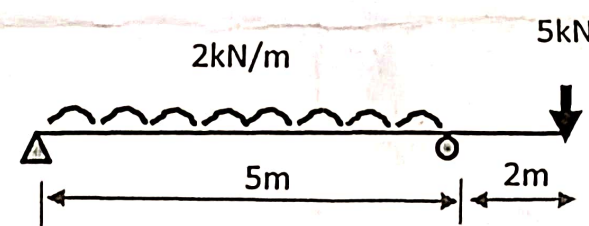
GROUP-A

Question number 1 is compulsory and attempt any two from the rest in the group -A

SERIAL NO.	BODY OF THE QUESTION	MARKS	Mapped CO
1.	Chose a structure out of Building, Bridge, Tower or any historically famous one. Draw a schematic diagram and a dimensional sketch with boundary conditions along with forces acting on it. Draw a FBD for analysis of that particular structure,	2.5	CO1
2.	Write a short note on the following with a neat FBD: a) Continuous girder beam of 4 spans supported over one fixed, two intermediate hinge and a roller support. The entire beam is subjected to UDL of 4kN/m. State with justification whether the girder is determinate or not. b) A cantilever 2d Truss with 5members, supported over one hinge support and one roller support. The truss is subjected to a vertically downward load of 10kN at the end. Also find force in any one member with nature. Chose suitable geometrical dimension of the Truss structure. c) Mention clearly on load specifications as per IS875 part2.	2+2+1	CO1
3.	Define Zero Force Member in case of Truss structure and explain its importance. Perform analysis of the TRUSS by method of sections and identify zero force members. 	5	CO2
4.	a) Distinguish among stable, unstable and over stable structures using equations and examples. b) Differentiate between static and kinematic determinacy, indeterminacy using equations and examples. c) For a single bay single storey PORTAL FRAME Structure with fixed supports, calculate the degree of static and kinematic indeterminacy.	2+2+1	CO2
5.	A simply supported over hanged beam is subjected to UDL of 3kN/m over its entire span. The simply supported span and over-hanged spans are 3m and 1.5m respectively. Draw FBD, SFD and BMD of the beam showing salient values of them on the diagram. Also calculate the point of contra flexure showing its location on the deflected shape of the loaded beam.	5	CO2

GROUP-B

(Answer all)

Question No.	Body of the Question	Marks	Mapped CO
6.	a) Draw the equivalent conjugate beam cantilever beam loaded with udl 'w' kN/m as shown in fig. 4 b) Hence, find the slope and deflection at free end. The beam has moment of inertia $2I$ and I, for the 1st half and last half span, respectively.  Fig.4	2+4	CO3
7.	Find the slope and deflection for the beam loaded as shown in fig. 5 below using any methods. EI is constant throughout the span.  Fig.5	5	CO4
8.	Write M-A theorem?	1.5	CO3

COURSE OUTCOMES:

CO1: Acquire the knowledge of structural systems, elements, joints, loads, stability, equilibrium, compatibility and indeterminacy.

CO2: Able to compute the internal forces in cable, arch, trusses, beams and frames.

CO3: Achieve the idea to apply geometric methods to obtain slope and deflections.

CO4: Gain the idea to apply Energy methods to obtain slope and deflections.

CO5: Evaluate and draw the influence lines diagrams for reactions, shears and bending moments in beams, girders due to moving load.

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CEC 401

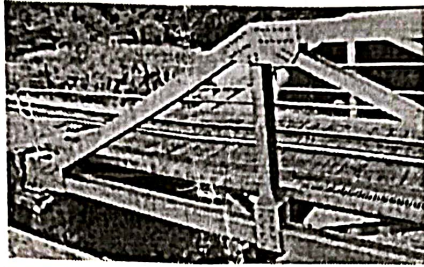
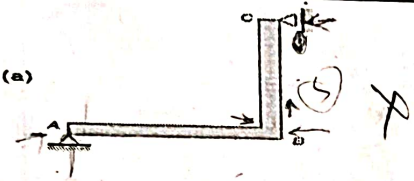
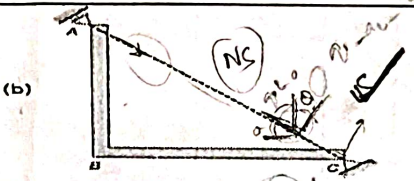
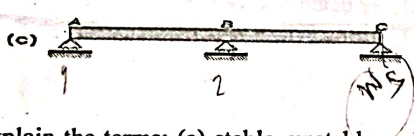
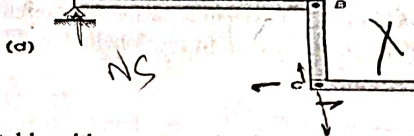
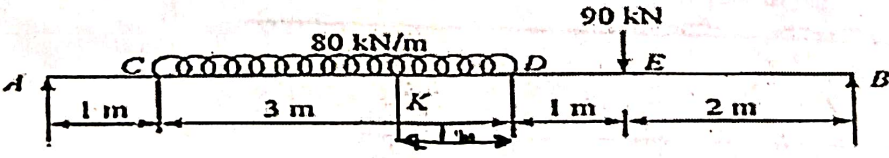
BTech/Regular/End-Sem/2025-26

FM: 60

TIME: Three Hrs.

GROUP-A

Attempt question number one and any two from the rest.

SERIAL NO.	BODY OF THE QUESTION	MARKS	Mapped COs
1.	A partial view of steel TRUSS bridge is shown below. Answer the following from analysis point of view: a) Type of truss and joints b) name and indicate the type of support c) indicate loading types and its application for the purpose of analysis d) indicate zero force member with proper explanation e) draw a simplified sketch for the truss for analysis and explain the methods for analysis using relevant equations. 	5x2=10	CO1
2.	(a)  (b)  (c)  (d)  Explain the terms: (a) stable, unstable and over stable with respect to load carrying systems as above. (b) Also explain each system for their condition as far as force transmission is concerned. Draw FBD in each case.	4+4x1.5=10	CO1
3.	Solve the following using ILD and compare the solution using conventional method. a) A point load of 10kN moves on a girder of span 20m from left to right. Calculate maximum SF and maximum BM at a section 5m from the left support. b) A uniformly distributed live load of 60 kN/m run of length 5 metres moves on a girder of span 16 metres. Find the maximum positive and negative shear force at a section 6 metres from the left end. Also find maximum BM.	2x5=10	CO5
4.	 Explain ILD and Muller- Breslau principle with the help of proper explanation. For the above loaded beam find the following using ILD concept: a) reaction forces b) SF at 'K' c) BM at D. Also validate each result with conventional method.	3+3+2+2	CO5
5.	Write a short note on the following: i) structure and mechanism ii) stiffness and flexibility iii) BIS code 1893 and 456 iv) rail bridge vs road bridge v) kinematic vs static indeterminacy of 2d and 3d structures.	5x2=10	CO1

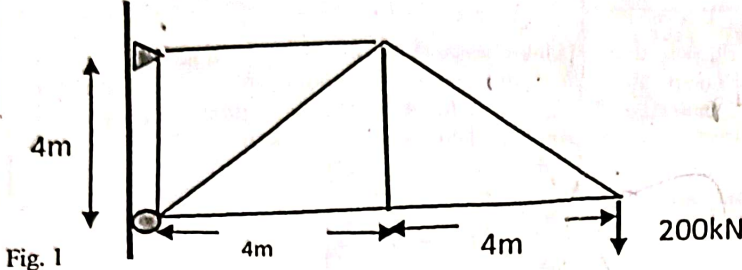
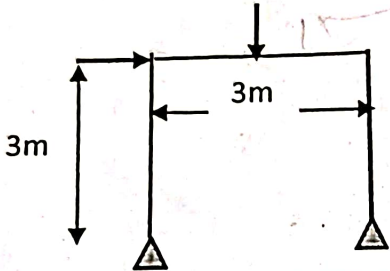
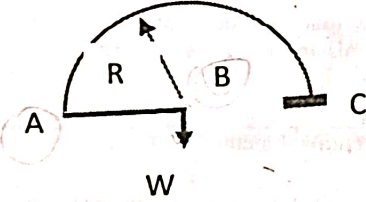
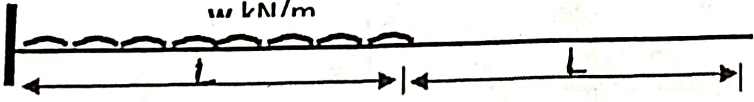
6-15 1300

$R_B = \frac{6 \times 3}{16}$

GROUP-B

Answer any three questions

BODY OF THE QUESTION

SERIAL NO.	BODY OF THE QUESTION	MARKS	COSs
6	For the plane truss as shown in the fig.1 below, find the displacement under 200 kN load. The cross sectional area 2000 mm², while $E = 200 \text{ kN/mm}^2$  Fig. 1	10	CO2
7.	Analyse the two hinge portal frame by SE method and draw the bending moment diagram. Two equal loads are acting as shown in fig.2 below. MI of the beam is twice that of column.  Fig. 2	10	CO4
8.	Find the free end vertical deflection for the semi circular cantilever beam loaded with a point load W at the free end 'B' as shown in fig. 3 below using energy methods. EI is constant throughout the span.  Fig. 3	10	CO3
9.	a) State and explain Maxwell reciprocal theorem. b) Find the deflection at free end of the cantilever beam using Maxwell reciprocal theorem.  Fig. 4	10	CO3

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